$$Sigmoid\ (S) = \frac{1}{1+e^{-x}};\ \ s' = s(1-s)$$

$$Sigmoid\ \big(\sigma(x)\big) = \frac{1}{1+e^{-x}};\ \ \sigma(x)' = \sigma(x)(1-\sigma(x))$$

$Chain\ rule$:

$$u = 2x + 5$$

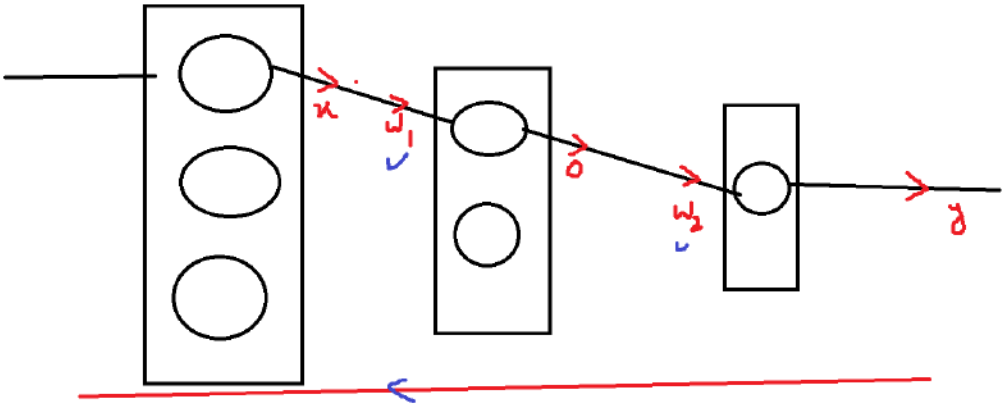
$$y = 3u + 4$$

$$\frac{dy}{dx} = in\ y\ equation\ x\ availabe : No$$

$$\frac{dy}{du}:\ in\ y\ equation\ u\ availabe : yes$$

$$\frac{du}{dx}:\ in\ u\ equation\ x\ availabe : yes$$

$$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx}$$



$$w_{1new} = w_{1old} - lr * \frac{dj}{dw_1}$$

$$w_{2new} = w_{2old} - lr * \frac{dj}{dw_2}$$

$$j = cost(y_{Ac} - y_p)$$

$$y_p = \sigma(w_2 * o + b_2)$$

$$j = interms\ of\ \sigma(w_2 * o + b_2)$$

$$\frac{dj}{dw_2} = availabe$$

$$\frac{dj}{dw_1} = not\ availabe$$

$$o = \sigma(w_1 * x + b_1)$$

$$j = cost(y_a - \sigma(w_2 * o + b_2))$$

$$\frac{dj}{dw_1} = \frac{dj}{d\sigma} * \frac{d\sigma}{do} * \frac{do}{dw1}$$

$$\frac{d\sigma}{do} = \sigma * (1 - \sigma)$$

$$case - 1: \ when\ \sigma = 0 \ the\ diff\ value\ is\ \sigma * (1 - \sigma) = 0$$

$$case - 2: \ when\ \sigma = 1 \ the\ diff\ value\ is\ \sigma * (1 - \sigma) = 0$$

$$case - 2: \ when\ \sigma = 0.5 \ the\ diff\ value\ is\ \sigma * (1 - \sigma) = 0.25$$

$$\sigma * (1 - \sigma)\ ranges\ from\ 0\ to\ 0.25$$

$$assume\ that\ \ \sigma * (1 - \sigma) = 0.25$$

$$\frac{dj}{dw_1} = \frac{dj}{d\sigma} * \frac{d\sigma}{do} * \frac{do}{dw1}$$

$$\frac{dj}{dw_1} = \frac{dj}{d\sigma} * 0.25 * \frac{do}{dw1} \qquad i - h - o$$

$$i - h - h - \ o \ : \ 0.25 * 0.25$$

$$i - h - h - h - \ o: 0.25 * 0.25 * 0.25 = 0.0001625$$

$$\frac{dj}{dw_1} = i - h - h - h - \ o \cong \ \ 0$$

- *when Neural network hiddenlayers increase the differentation of activation number increase*
- *every differentation of sigmoid activation function has* max *value* 0.25
- *when we do more multiplies of* 0.25 *at one point of time the value will be apprx* 0
- *assume* 100 *hidden layers* : $(0.25)^{100} \cong 0$

$$w_{1new} = w_{1old} - lr * 0$$

$$w_{1new} = w_{1old}$$

- *weights not updating*
- *no learning*
- *but still error is maximum*
- *this problem is called Vanish Gardients problem*
- *This happens becuase hidden layer neurons are using sigmoid and* tanh *activation function*
- *So Hidden layers will not use sigmoid* tanh *and softmax activation functions*

- *sigmoid , softmax and* tanh *all are facing Vanish gardients problem*

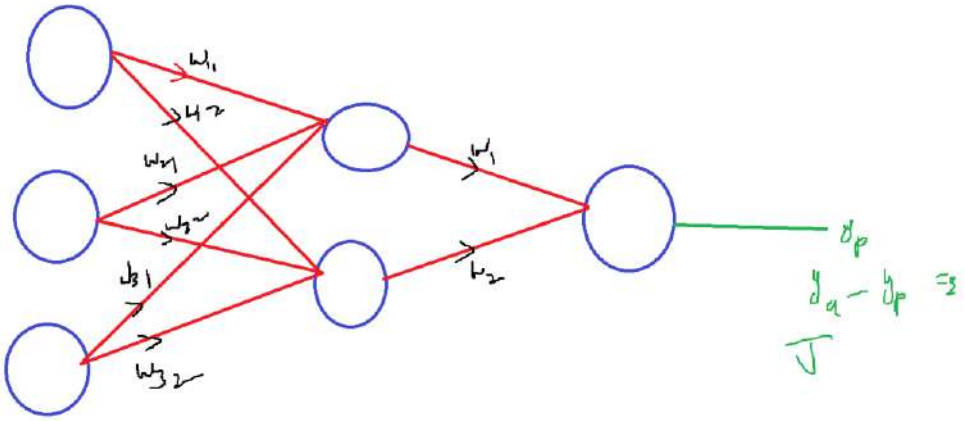
- $sigmoid\ wil\ use\ in\ output\ layer\ bi\ classification$
- $softmax\ wil\ use\ in\ output\ layer\ multi\ classification$
- tanh $-1\ to\ 1$ $if\ any\ use\ cases\ we\ need\ output\ -1\ to\ 1\ then\ go\ for\ tanh$
- $before\ ReLU\ introduces$ tanh $used\ in\ hidden\ layers\ also$
- $Still\ LSTM\ RNN\ AutoEncoders\ are\ using$ tanh $in\ hidden\ layers$
- $ReLU$ 0 to inf $are\ used\ in\ hidden\ layers$
    - $ReLU\ assumes\ if\ inputs\ are\ negative\ it\ assmue\ as\ zero$
    - $What\ if\ all\ inputs\ are\ negative\ then\ ReLU\ converts\ all\ inputs\ into$ 0
    - $in\ this\ situation\ we\ will\ use\ Leaky\ ReLU$

$GPT\ models\ we\ use\ GELU\ in\ Hidden\ layers$: $Guassian\ Error\ Linear\ unit$

$In\ output\ layer\ no\ activation\ ,just\ use\ raw\ values\ that\ logits$

$$\sigma(w_1 * x + b) = activation\ on\ logits$$

$$w_1 * x + b = logit$$


w_{11}
w_{12}
w_{21}
w_{22}
w_{31}
w_{32}
w'_1
w'_2
y_p
$y_a - y_p$ =
J